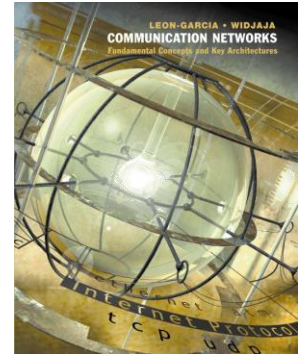
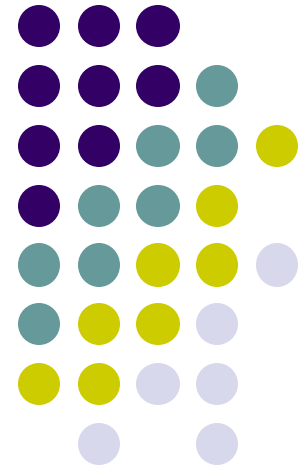


Lecture 5(Chapter 8)

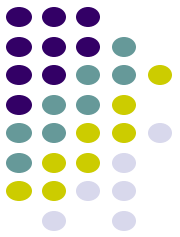
Communication Networks and Services



Internet Routing Protocols

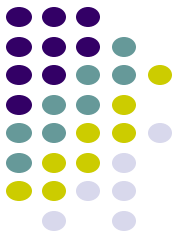


Outline

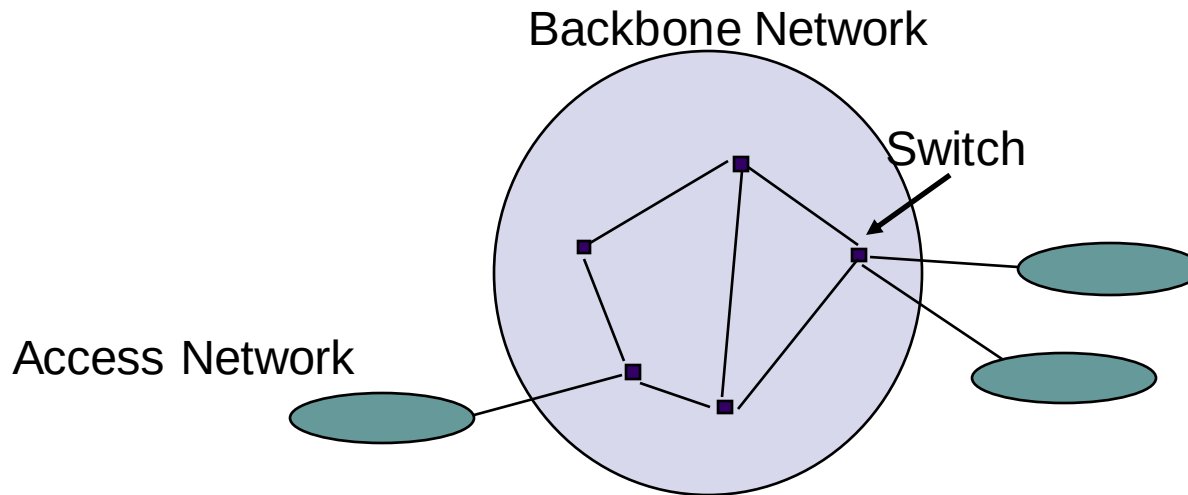


- Datagrams *and Virtual Circuits*
- Basic Routing
- Routing Information Protocol (RIP)
- Open Shortest Path First (OSPF)
- Border Gateway Protocol (BGP)

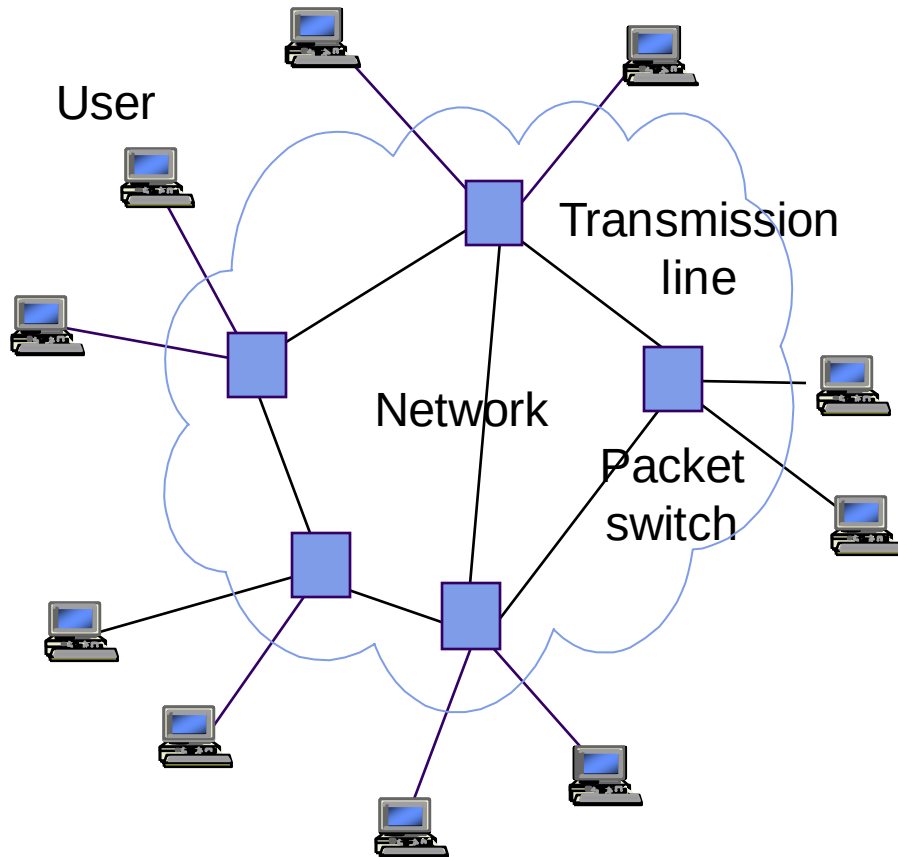
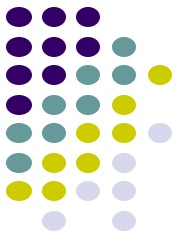
The Switching Function



- Dynamic interconnection of inputs to outputs
- Enables dynamic sharing of transmission resource
- Two fundamental approaches:
 - Connectionless
 - Connection-Oriented: Call setup control, Connection control



Packet Switching Network



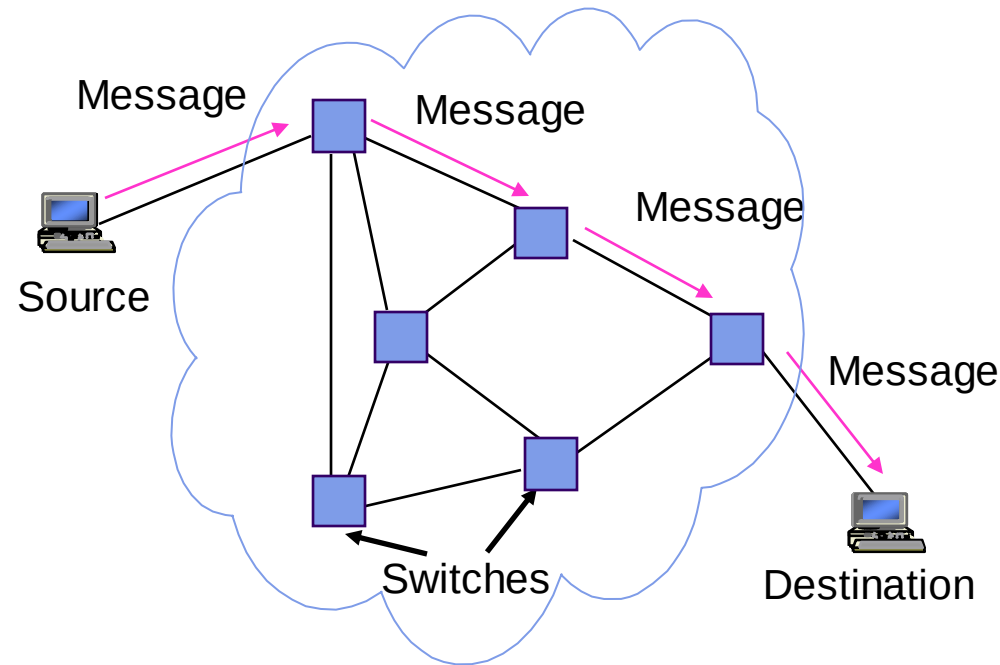
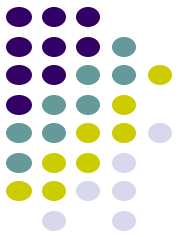
Packet switching network

- Transfers packets between users
- Transmission lines + packet switches (routers)
- Origin in message switching

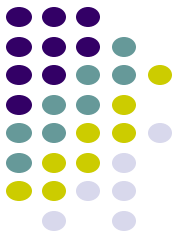
Two modes of operation:

- Connectionless — sms
- Virtual Circuit — mobile voice call

Message Switching

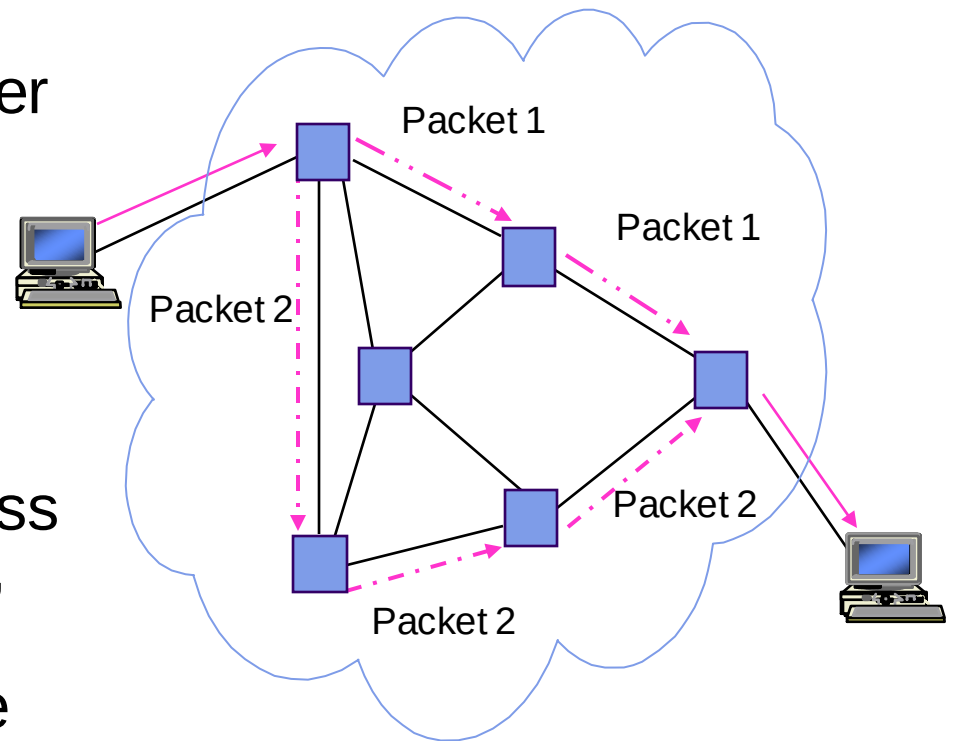


- Message switching invented for telegraphy
- Entire messages multiplexed onto shared lines, stored & forwarded
- Headers for source & destination addresses
- Routing at message switches
- Connectionless

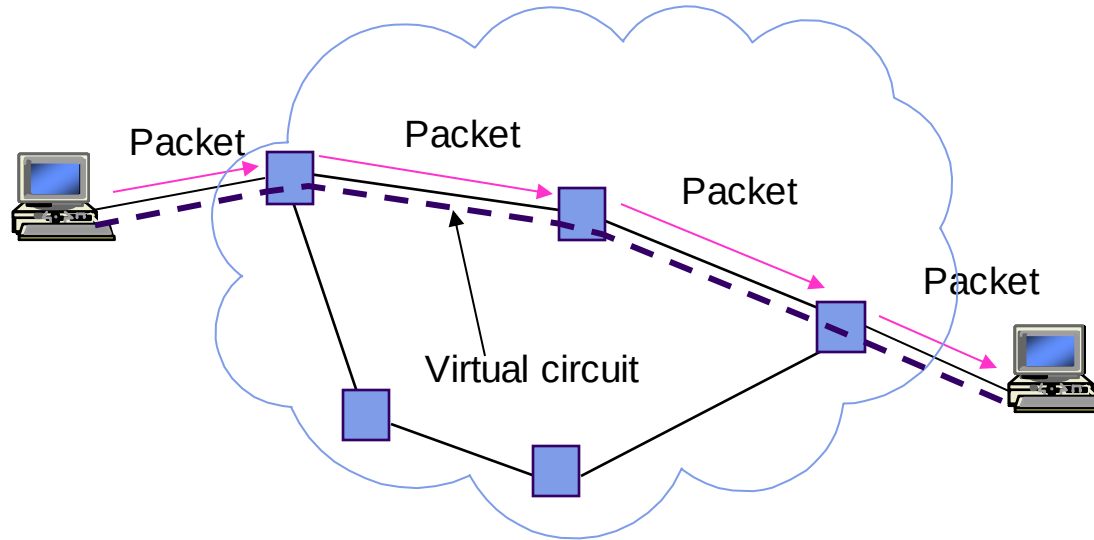
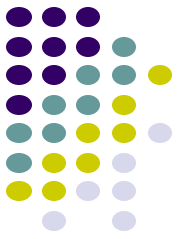


Packet Switching - Datagram

- Messages broken into smaller units (packets)
- Source & destination addresses in packet header
- Connectionless, packets routed independently (datagram)
- Packet may arrive out of order
- Pipelining of packets across network can reduce delay, increase throughput
- Lower delay than message switching, suitable for interactive traffic

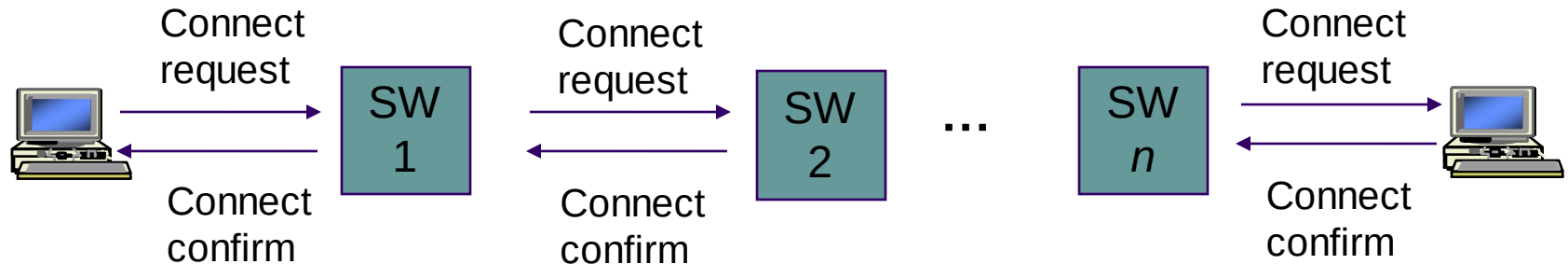
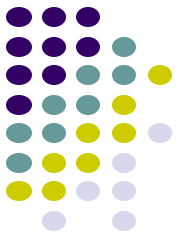


Packet Switching – Virtual Circuit



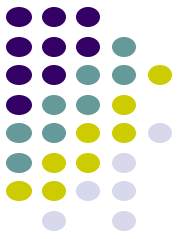
- Call set-up phase sets up pointers in fixed path along network
- All packets for a connection follow the same path
- Abbreviated header identifies connection on each link
- Packets queue for transmission
- Variable bit rates possible, negotiated during call set-up
- Delays variable, cannot be less than circuit switching

Connection Setup

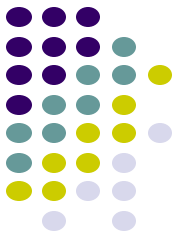


- Signaling messages propagate as route is selected
- Signaling messages identify connection and setup tables in switches
- Typically a connection is identified by a local tag, Virtual Circuit Identifier (VCI)
- Each switch only needs to know how to relate an incoming tag in one input to an outgoing tag in the corresponding output
- Once tables are setup, packets can flow along path

Outline



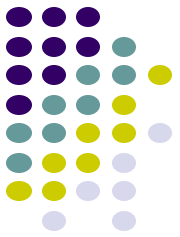
- Datagrams *and Virtual Circuits*
- Basic Routing
- Routing Information Protocol (RIP)
- Open Shortest Path First (OSPF)
- Border Gateway Protocol (BGP)



Routing and Forwarding

- Routing
 - How to determine the routing table entries
 - carried out by routing daemon
- Forwarding
 - Look up routing table & forward packet from input to output port
 - carried out by IP layer

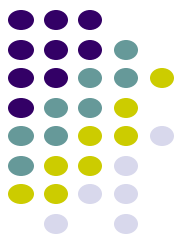
Routers exchange information using routing protocols to develop the routing tables



Routing Table Entries

- Destination IP Address:
 - complete host address or network address
- IP address of
 - next-hop router or directly connected network
- Flags
 - Is destination IP address a net address or host address?
 - Is next hop, a router or directly connected?
- Network interface on which to send packet

Static routing



- Used on hosts or on very small networks
- Manually tell the machine where to send the packets for each prefix

```
% netstat -nr
```

Routing Table:

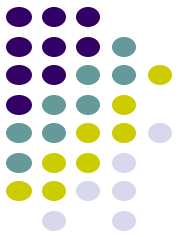
Destination Interface	Gateway	Flags	Ref	Use

-				
127.0.0.1	127.0.0.1	UH	0	0 100
128.100.10.0	128.100.10.9	U	3	548 1e0
224.0.0.0	128.100.10.9	U	3	0 1e0
default	128.100.10.2	UG	0	35792

U-Route is up H-route is to host (else route is to network)

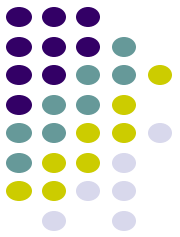
G-route to gateway (else direct connection)

Forwarding Procedure



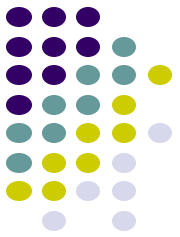
- Does routing table have entry that matches complete destination IP address? If so, use this entry to forward
- Else, does routing table have entry that matches the longest prefix of the destination IP address? If so, use this entry to forward
- Else, does the routing table have a default entry? If so, use this entry.
- Else, packet is undeliverable

Autonomous Systems



- Global Internet viewed as collection of autonomous systems.
- **Autonomous system (AS)** is a set of routers or networks administered by a single organization
- Same routing protocol need not be run within the AS
- But, to the outside world, an AS should present a *consistent picture of what ASs are reachable* through it
- **Stub AS:** has only a single connection to the outside world.
- **Multihomed AS:** has multiple connections to the outside world but refuse to carry transit traffic.
- **Transit AS:** has multiple connections to the outside world, and can carry transit and local traffic.

Inter and Intra Domain Routing



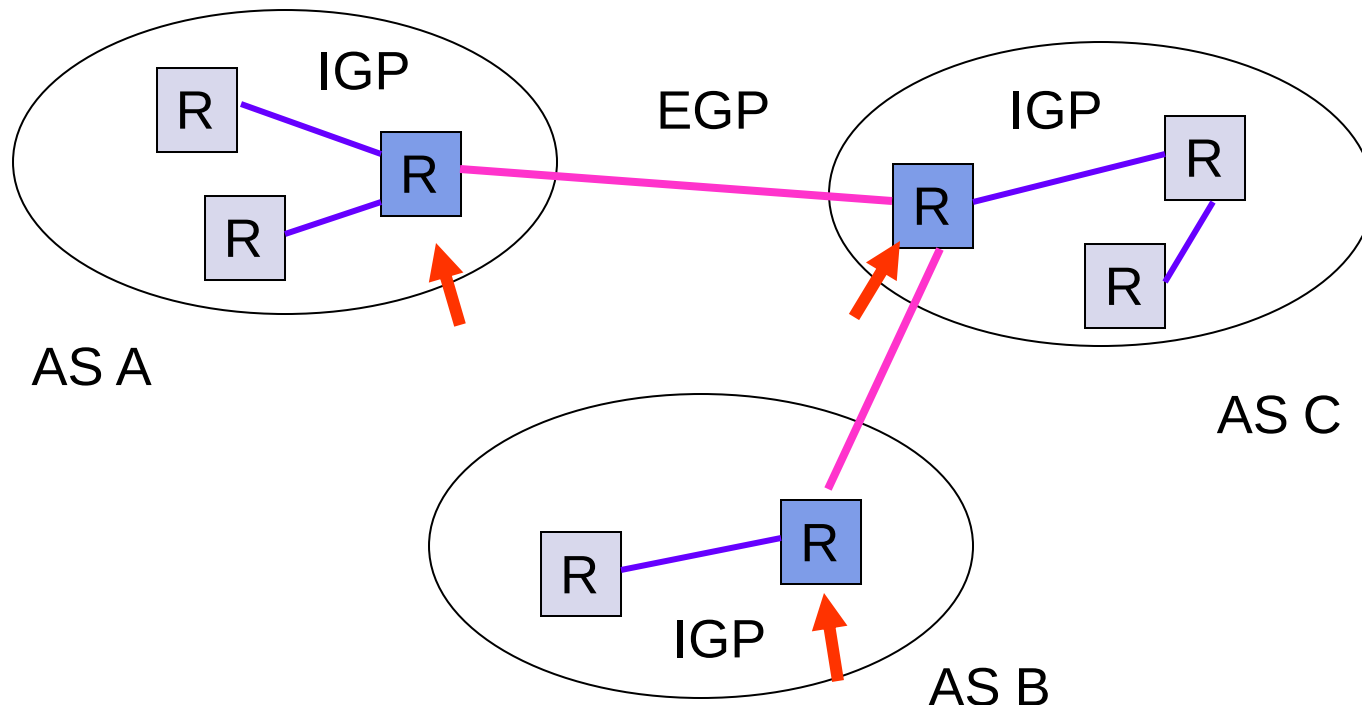
Interior Gateway Protocol (IGP): routing within AS

- RIP, OSPF

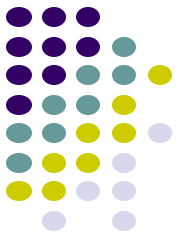
Exterior Gateway Protocol (EGP): routing between AS's

- BGPv4

Border Gateways perform IGP & EGP routing

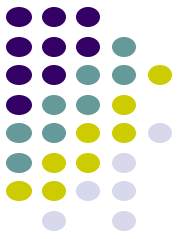


Outline



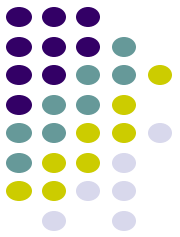
- Datagrams *and Virtual Circuits*
- Basic Routing
- Routing Information Protocol (RIP)
- Open Shortest Path First (OSPF)
- Border Gateway Protocol (BGP)

Routing Information Protocol (RIP)



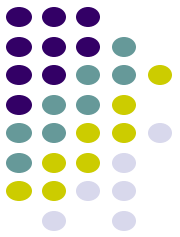
- RFC 1058
- **RIP** based on routed, “route d”, distributed in BSD UNIX
- Uses the **distance-vector algorithm**
- Runs on top of UDP, port number 520
- Metric: number of hops
- Max limited to 15
 - suitable for small networks (local area environments)
 - value of 16 is reserved to represent infinity
 - small number limits the *count-to-infinity* problem

RIP Operation



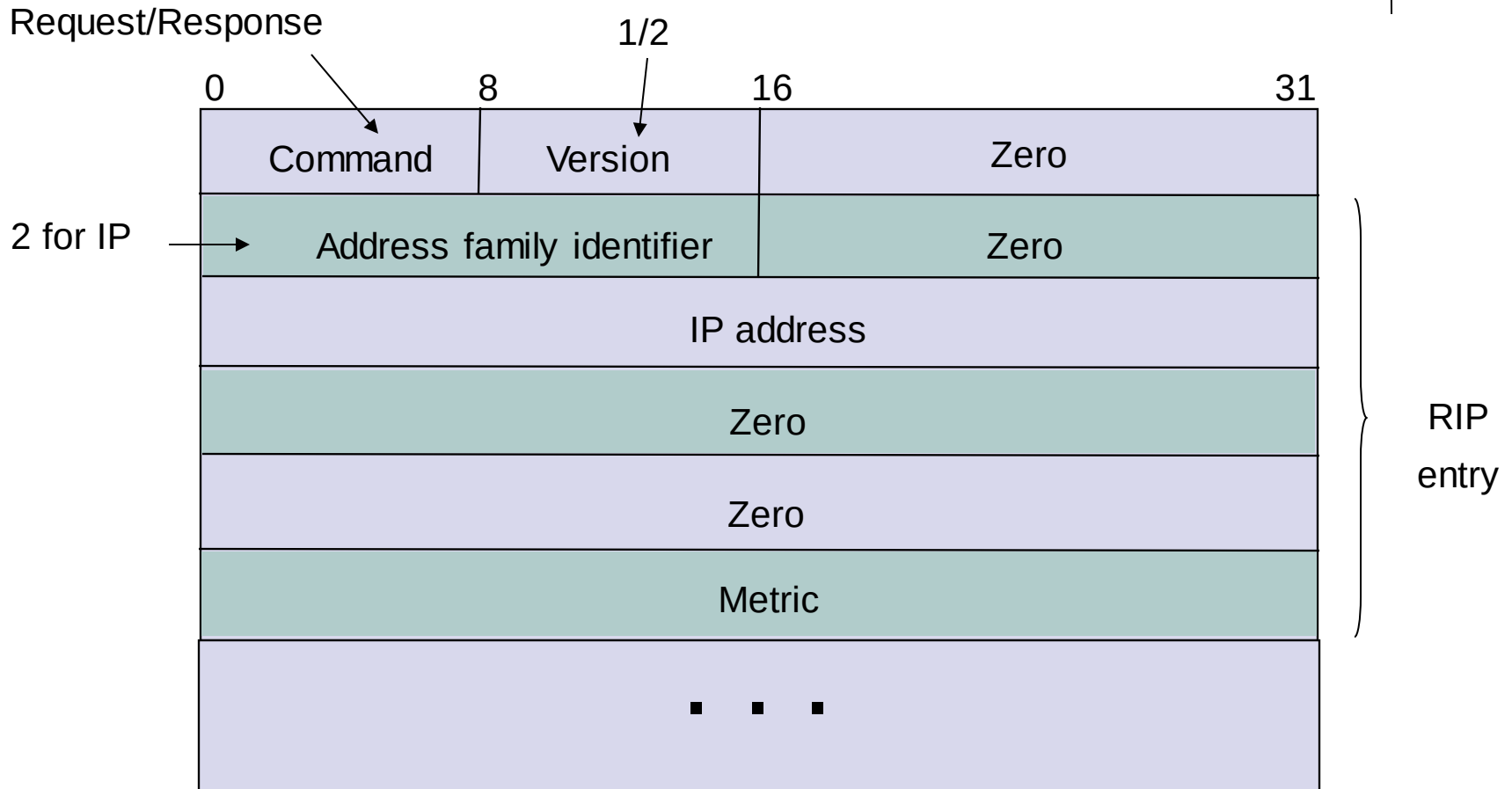
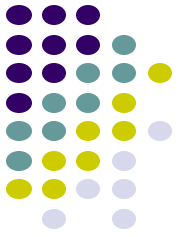
- Router sends update message to neighbors every 30 sec
- A router expects to receive an update message from each of its neighbors within 180 seconds in the worst case
- If router does not receive update message from neighbor X within this limit, it assumes the link to X has failed and sets the corresponding minimum cost to 16 (infinity)
- Convergence speeded up by triggered updates
 - neighbors notified immediately of changes in distance vector table

RIP Protocol



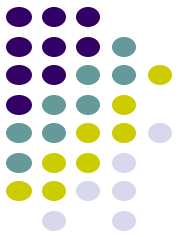
- Routers run RIP in active mode (advertise distance vector tables)
- Hosts can run RIP in passive mode (update distance vector tables, but do not advertise)
- RIP datagrams broadcast over LANs & specifically addressed on pt-pt or multi-access non-broadcast nets
- Two RIP packet types:
 - *request* to ask neighbor for distance vector table
 - *response* to advertise distance vector table
 - periodically; in response to request; triggered

RIP Message Format



Up to 25 RIP entries per message

RIP Message Format

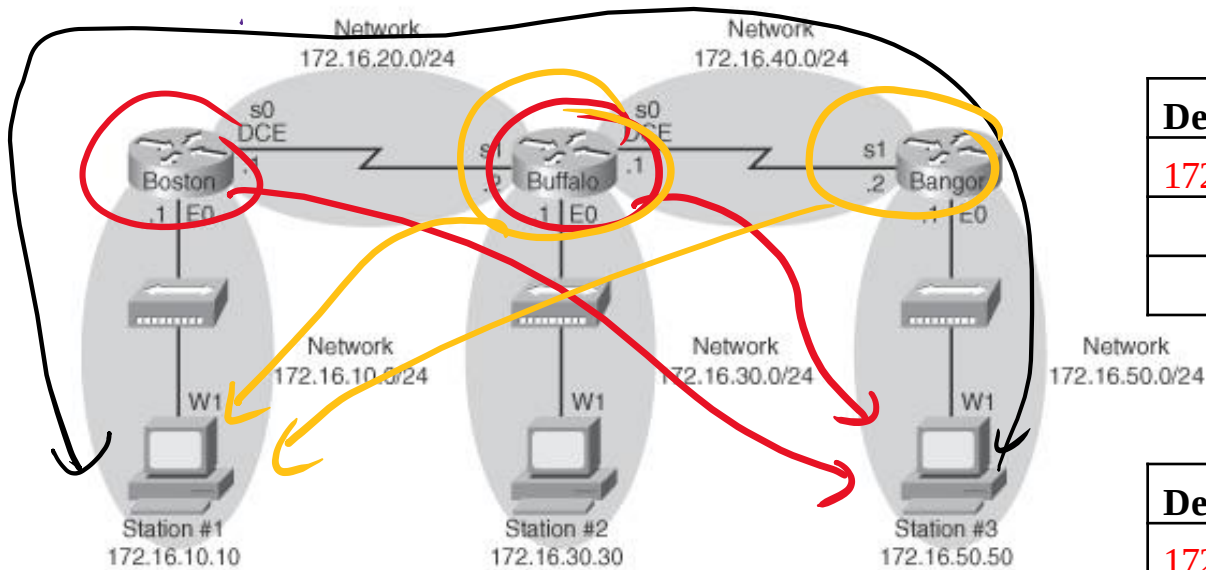
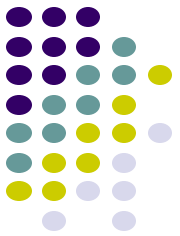


- Command: request or response
- Version: v1 or v2
- One or more of:
 - Address Family: 2 for IP
 - IP Address: network or host destination
 - Metric: number of hops to destination
- Does not have access to subnet mask information
- Cannot work with variable-length subnet masks

RIP v2 (RFC 2453):

- Subnet mask, next hop, routing domain
- can work with CIDR
- still uses max cost of 16

Static Route Exccercize



Boston R

Destination	Gateway	Preference
172.16.50.0/24	172.16.20.2	60

Buffalo R

Destination	Gateway	Preference
172.16.50.0/24	172.16.40.2	60
172.16.10.0/24	172.16.20.1	60

Bangor R

Destination	Gateway	Preference
172.16.10.0/24	172.16.40.1	60

from st 1 → st 3

from st 3 → st 1